AML-3406

AL and ML Capstone Project

Capstone Project Weekly Progress Report

Semester	Fall-2024 Semester 3rd
Course Code	AML-3406
Section	Section 1
Project Title	Car Damage Detection
Group Name	Group B
Student names/Student IDs	Sakshi (C0908000) – TL Bansil Patel (C0912873) Harsh Mohile (C0912872) Meet Patel (C0910378) Rachit Bhatt (C0902810)
Reporting Week	Week 12
Faculty Supervisor	William Pourmajidi

1. Grade the level of collaboration from different aspects between team members: (use: good, medium, below expectation)

Sakshi	Bansil Patel	Harsh Mohile	Meet Patel	Rachit Bhatt
Good	Good	Good	Good	Good

2. Tasks outlined in the previous Weekly Progress Report for this reporting week at the individual level and group level

2.1 Individual tasks (each team member)

Sakshi: Enhance the GitHub Actions pipeline by adding real-time monitoring and alerting features to track deployment status.

Bansil Patel & Meet Patel: Plan to expand testing with more diverse datasets, focusing on rare car damage types for improved model generalization**.**

Harsh Mohile & Rachit Bhatt: Further optimize the Deployment UI to improve user experience and make it more intuitive for end users.

2.2 Your team's tasks

- Model Enhancements: Plan to explore advanced model architectures to improve accuracy, aiming for higher performance in detecting complex car damages.
- Deployment Pipeline: Add real-time monitoring to the deployment pipeline, enabling better tracking of deployment success and quick rollback if needed.
- UI Optimization: Improve the Deployment UI with new features such as interactive visualizations of model predictions and user feedback integration.
- Scalability: Focus on optimizing the model's performance for large-scale data processing to handle a wider variety of car images in real-world applications.

3. Progress made in Reporting Week at an individual level and group level

3.1 Individual progress

Sakshi: Resolved repository configuration issues to ensure consistent deployments.

Bansil Patel & Meet Patel: Validated the model's robustness by testing on multiple datasets, confirming its generalization to different car damage types.

Harsh Mohile & Rachit Bhatt: Conducted extensive testing across staging and production environments, ensuring reliable performance.

3.2 Your team's progress

- Finalized the automated deployment pipeline with GitHub Actions, ensuring smooth deployments across all environments. Fully integrated the UI with the backend, ensuring seamless deployment in staging and production.

4. The areas/tasks you could not make progress and/or complete as scheduled or the difficulties encountered in this reporting week at individual level and group level.

4.1 Individual project blockers

Sakshi: Minor integration issues were resolved by refining repository configurations.
Bansil Patel & Meet Patel: Delays in backend-UI communication but resolved during testing.
Harsh Mohile & Rachit Bhatt: N/A.

4.2Your team's project blockers

- Multi-branch setup caused delays, resolved with configuration adjustments.
- Minor adjustments to align the UI with production configurations.

5. Tasks to be completed in the next week at the individual level and group level

5.1 Individual tasks

Sakshi, Bansil Patel & Meet Patel: N/A

Harsh Mohile & Rachit Bhatt: N/A

5.2 Your team's tasks

The project has been completed successfully.

6. Include the tasks from your sprint planning (GitHub/ZenHub) for the present period.

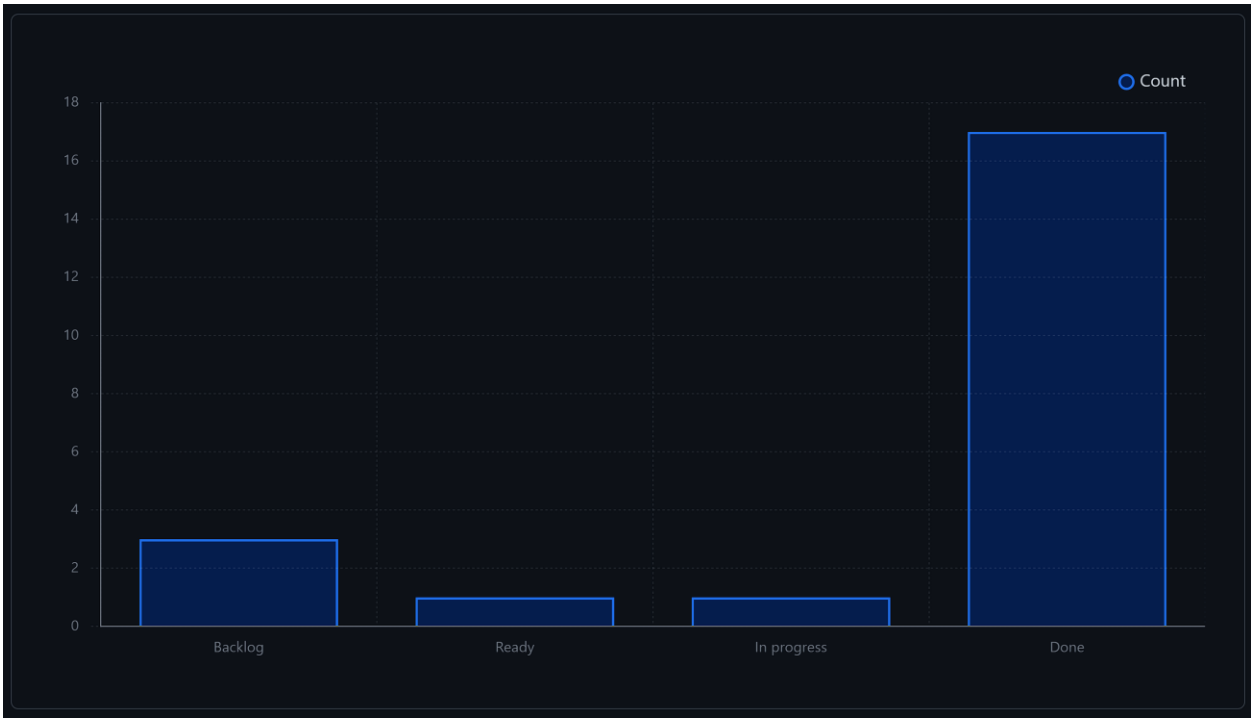

Figure 1: Weekly Activities

- In this week, there were a lot of issues resolved from the backlog, contributing to development.

7. Include charts/graphs (e.g., burn down charts) from your project management tool (Github/Zenhub) that shows your progress for the period of this report.

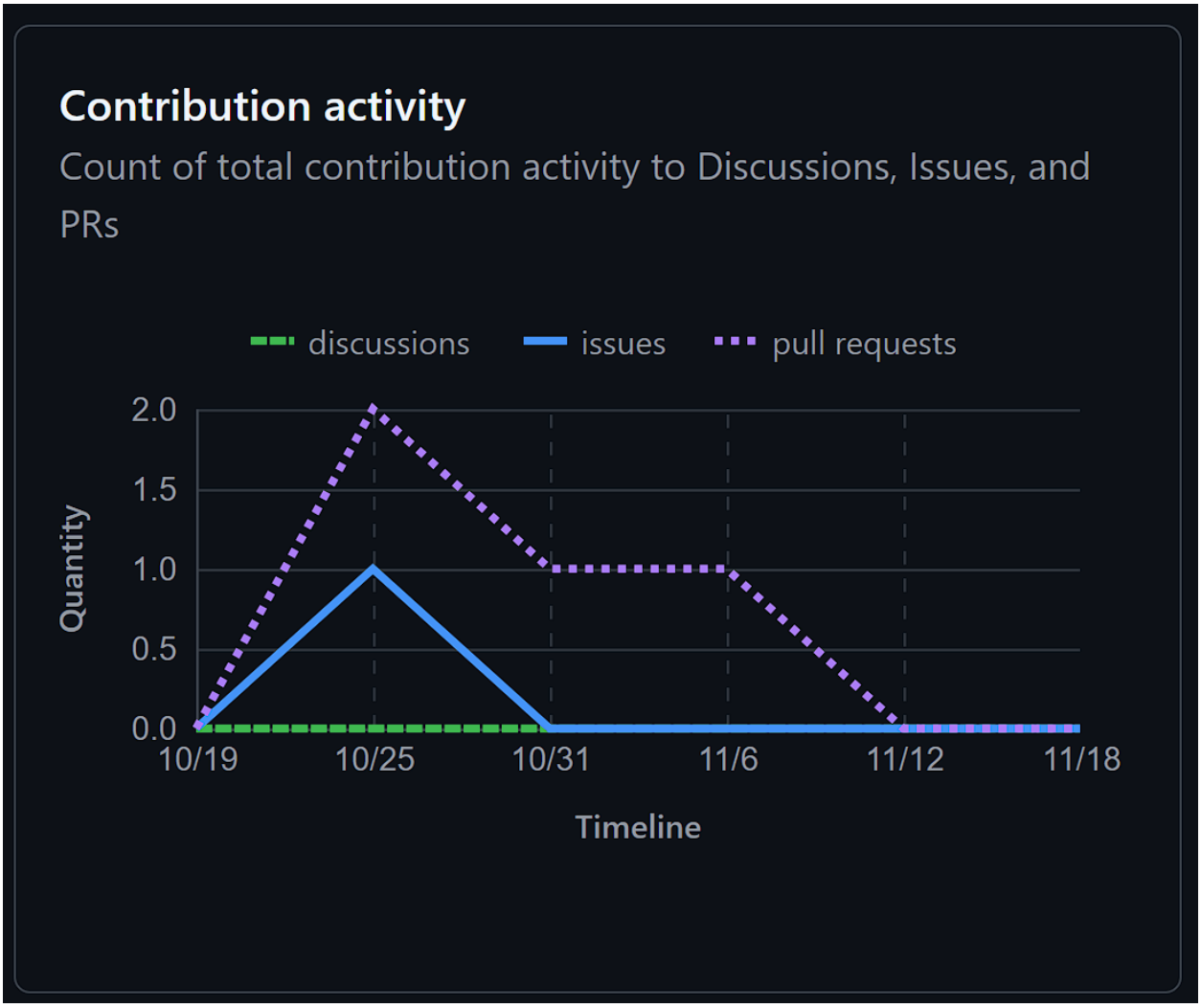

Figure 2: Contributors in the Week

- This week, we had a few PRs and issues resolved as compared to previous week, demonstrating team hard work and dedication.

8. **Include a note and address to your project GitHub with list of codes uploaded/updated on GitHub in this reporting week.**

GitHub: https://www.github.com/rachit-bhatt/Capstone-Project